

Gauge Neural Dynamics: Learning Context-Dependent Coordinate Systems in Biological Neural Representations

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Abstract

Place fields rotate with a cue card. Grid phases shift coherently between environments. Motor cortical trajectories for different reach directions look like rotated copies of one another. In each case the geometry appears to be the same object seen in a different frame, but fitting one manifold to pooled activity has no way to say so and absorbs the difference into the residual. We model the frame change directly: an invariant latent state z is read out through a tuning map f shared across contexts, after a context-dependent transformation $x_c = f(T_c(z))$, with T_c the exponential of a context-dependent element of a Lie algebra learned from data. Identity and exact invertibility follow from the construction; closure under composition does not, so we measure it, using structure constants obtained by projecting the learned commutators back onto the algebra. We define four metrics for context-dependent geometry and apply them identically to our model and to seven baselines. On simulations of hippocampal remapping, grid-cell phase translation and a trained reaching network, the learned gauge recovers the generating transformations and predicts activity in contexts held out of training, which no baseline does. The learned algebra is abelian where the biology says it should be and non-abelian where it should not. We also show that a single-chart gauge field cannot cover a cyclic context variable, the failure appearing where the topology predicts. Two conditions were built so that no latent gauge transformation exists; the model fails on both, as intended.

1 Introduction

Rotate the polarising cue in a cylinder and the entire CA1 place-cell map rotates with it (Muller and Kubie, 1987; Bostock et al., 1991). Stretch the enclosure and the map stretches (O’Keefe and Burgess, 1996). Morph between two familiar shapes and it deforms in a graded way (Wills et al., 2005; Leutgeb et al., 2005). In entorhinal cortex, grid modules shift and rotate coherently across environments (Fyhn et al., 2007) and rescale after the enclosure changes (Barry et al., 2007). In motor cortex, the rotational structure of the population response differs systematically with reach condition (Churchland et al., 2012; Vyas et al., 2020), and yet the geometry is stable enough across days, and even across animals, to be recovered by alignment (Gallego et al., 2020; Degenhart et al., 2020; Safaie et al., 2023).

These observations share a form. The manifold does not appear to change shape; it appears to be expressed in a different coordinate system. Standard latent variable analysis cannot represent that. A single embedding is fitted to activity pooled across conditions (Cunningham and Yu, 2014; Gallego et al., 2017), so a coherent rotation of the map and an incoherent scattering of it are, to the model, the same amount of unexplained variance.

Gauge theory is the natural language for a quantity defined only up to a local choice of coordinates, together with a rule for translating between choices (Kobayashi and Nomizu, 1963; Nakahara, 2003). We use it literally. Let $z \in \mathcal{M}$ be the latent computational state, let f be the tuning of the recorded

population, and let each context c act on the frame:

$$x_c = f(T_c(z)), \quad T_c : \mathcal{M} \rightarrow \mathcal{M}. \quad (1)$$

The content of (1) lies in what it forbids. Since f is shared and never sees the context, any difference between contexts must be a transformation of the latent. Effects that are not of that form cannot be absorbed and appear as error instead. Two of the conditions below are constructed so that no latent gauge transformation exists, and we check that the model fails on them.

We contribute, first, a framework in which T_c is the exponential of a context-dependent element of a learned finite-dimensional Lie algebra (Sec. 3), with identity and exact invertibility structural and closure measured rather than assumed, together with a proof that a single-chart field of this form cannot cover a context space on which the required group elements wind (Prop. 1). Second, three simulations built from established phenomenology – place-cell remapping, grid-cell translation on a torus, and a reaching RNN trained by BPTT – on which the learned transformations recover the generating geometry (Sec. 5); one consequence, that the eigenvalues of latent dynamics are context invariant while their eigenplanes rotate, is testable in existing recordings without fitting our model. Third, four metrics for representational consistency across contexts, each with the diagnostic needed to keep it from being satisfied trivially (Sec. 4).

2 Related work

Population activity is routinely summarised with latent variable models, from linear factorisations (Cunningham and Yu, 2014) to sequential autoencoders (Pandarinath et al., 2018) and contrastive embeddings (Schneider et al., 2023). Context and session changes, when handled, are handled post hoc: canonical correlation (Hotelling, 1936), orthogonal Procrustes (Schönemann, 1966), or semi-supervised manifold alignment (Ham et al., 2005), as used to stabilise brain–computer interfaces (Degenhart et al., 2020) and to compare across days and animals (Gallego et al., 2020; Safaie et al., 2023). All fit one transformation per recorded condition, so they cannot ask whether the transformations form a group and are undefined at a condition that was not recorded. Both questions are central here.

The phenomena we simulate are coherent place-field remapping (Muller and Kubie, 1987; Colgin et al., 2008), grid translation and rotation (Hafting et al., 2005; Fyhn et al., 2007; Stensola et al., 2012) and grid rescaling (Barry et al., 2007); that a single grid module’s manifold is a torus was shown with persistent homology by Gardner et al. (2022), which our second experiment mirrors. Network models treat remapping as attractor reorganisation (Burak and Fiete, 2009; Low et al., 2023) or as inference over a structural code (Whittington et al., 2020), where we ask instead what transformation relates the observed representations. Equivariant networks assume a known group (Cohen and Welling, 2016; Weiler and Cesa, 2019; Bronstein et al., 2017), and gauge equivariant networks handle the case where no global frame exists (Cohen et al., 2019; Bronstein et al., 2021). Nearest to us is work that learns a Lie algebra rather than assuming one (Dehmamy et al., 2021; Finzi et al., 2021; Connor et al., 2021; Keller and Welling, 2021); the difference is in the target, since there the group is a prior to impose and here it is a property of a biological system to be estimated and reported with its defects. Our topology term follows Moor et al. (2020), and the analysis uses persistent homology (Edelsbrunner et al., 2002; Carlsson, 2009; Tralie et al., 2018), which has an established role in neural coding (Singh et al., 2008; Curto, 2017; Chaudhuri et al., 2019).

3 Method

3.1 A learned Lie algebra of gauge transformations

Let $z \in \mathcal{M} \subseteq \mathbb{R}^d$ and let $c \in \mathcal{C}$ be an observable context variable: an environment label, a cue direction, a task condition. We learn generators $G_1, \dots, G_K \in \mathfrak{gl}(d)$ spanning $\mathfrak{g} = \text{span}\{G_k\}$ together with a context embedding $\theta : \mathcal{C} \rightarrow \mathbb{R}^K$, and set

$$A(c) = \sum_{k=1}^K \theta_k(c) G_k, \quad T_c(z) = \exp(A(c))z. \quad (2)$$

Three properties come for free. Smoothness of $c \mapsto T_c$ follows from that of θ and $\exp$; invertibility is exact and in closed form, $T_c^{-1} = \exp(-A(c))$, with no learned inverse; and $T_{c_{\text{ref}}} = I$ is enforced by anchoring, $\theta(c) \mapsto \theta(c) - \theta(c_{\text{ref}})$. Anchoring adds no capacity, since (1) is invariant under $z \mapsto Sz$, $T_c \mapsto ST_c S^{-1}$, $f \mapsto f \circ S^{-1}$, and anchoring merely fixes the residual left-multiplication freedom. Generators are normalised to unit Frobenius norm, removing the $\theta \leftrightarrow G$ scale redundancy so the defects below are comparable across runs.

The fourth property, closure under composition, is the informative one. By Baker–Campbell–Hausdorff, $\exp(A)\exp(B) = \exp(A + B + \frac{1}{2}[A, B] + \dots)$, so $\{T_c\}$ is closed if and only if $\mathfrak{g}$ is a Lie subalgebra. We do not impose this. Projecting the commutators back onto $\mathfrak{g}$ by least squares gives structure constants and a residual,

$$[G_i, G_j] = \sum_k f_{ij}^k G_k + R_{ij}, \quad \delta_{\text{closure}} = \langle \|R_{ij}\|_F / \|[G_i, G_j]\|_F \rangle_{i \neq j}, \quad (3)$$

averaged over pairs whose commutator is not numerically zero. The f_{ij}^k define a composition law on the algebra, $\theta_{a \cdot b} = \text{BCH}(\theta_a, \theta_b)$, truncated at second or third order. Both the training objective and the gauge consistency score are built on it, and it is exact when the learned algebra closes.

Writing $g_{ab} = T_b T_a^{-1}$ makes (1) look like a fibre bundle over $\mathcal{C}$, but the cocycle condition $g_{bc} g_{ab} = g_{ac}$ is then automatic, the transitions being built from one globally defined family, and so tests nothing. What is not automatic is whether the composite stays inside the family, which is what (3) measures. Curvature does become measurable once transitions are chained around a loop in context space, and we report the resulting holonomy.

3.2 An obstruction

Continuous context variables expose a limitation that discrete labels hide.

Proposition 1 (winding obstruction). *Let $\theta : \mathcal{C} \rightarrow \mathbb{R}^K$ be continuous and $T_c = \exp(\sum_k \theta_k(c) G_k) \in \text{GL}(d, \mathbb{R})^+$. For any loop $\gamma : S^1 \rightarrow \mathcal{C}$, the induced loop $t \mapsto T_{\gamma(t)}$ is null-homotopic in $\text{GL}(d, \mathbb{R})^+$. Hence if the ground-truth family restricted to γ represents a non-trivial class in $\pi_1(\text{GL}(d, \mathbb{R})^+) \cong \pi_1(\text{SO}(d))$, which is $\mathbb{Z}$ for $d = 2$ and $\mathbb{Z}/2$ for $d \geq 3$, no choice of K and no choice of generators can represent it.*

Proof. $\theta \circ \gamma$ is null-homotopic because $\mathbb{R}^K$ is contractible; let H contract it to a constant. Then $\exp(\sum_k H_k(\cdot, s) G_k)$ contracts $t \mapsto T_{\gamma(t)}$ within $\text{GL}(d, \mathbb{R})^+$, since $\exp$ is continuous and lands there. A null-homotopic loop represents the trivial class. $\square$

This is a statement about topology, not about optimisation or capacity, and the remedy is the usual one: pass to a covering of $\mathcal{C}$, or equivalently use more than one chart. Section 5.4 checks both directions.

3.3 Non-linear gauges and shared latent dynamics

Replacing matrices with vector fields extends (2) to diffeomorphisms. With $V_k(z) = G_k z + v_k + \eta m_k(z)$ for a small MLP m_k , we take T_c to be the time-one flow of $\sum_k \theta_k(c) V_k$, integrated with RK4 and inverted by integrating the negated field. At $\eta = 0$ this reduces to (2) exactly. The bracket $[V_i, V_j] = DV_j V_i - DV_i V_j$ is evaluated by automatic differentiation and projected pointwise onto $\text{span}\{V_k\}$, leaving (3) unchanged.

If the latent computation is shared, so is its flow. With canonical dynamics $\dot{z} = F(z)$, the dynamics measured in context c are the push-forward $F_c(w) = DT_c(T_c^{-1}w)F(T_c^{-1}w)$, which for a linear gauge is conjugation,

$$F_c = M_c F M_c^{-1}, \quad M_c = \exp(A(c)). \quad (4)$$

Conjugation preserves eigenvalues, so (4) predicts that the rotation frequency of the linearised latent dynamics is shared across conditions while the rotation plane turns with them. This can be checked in recorded data without fitting our model, and we check it in Sec. 5.3.

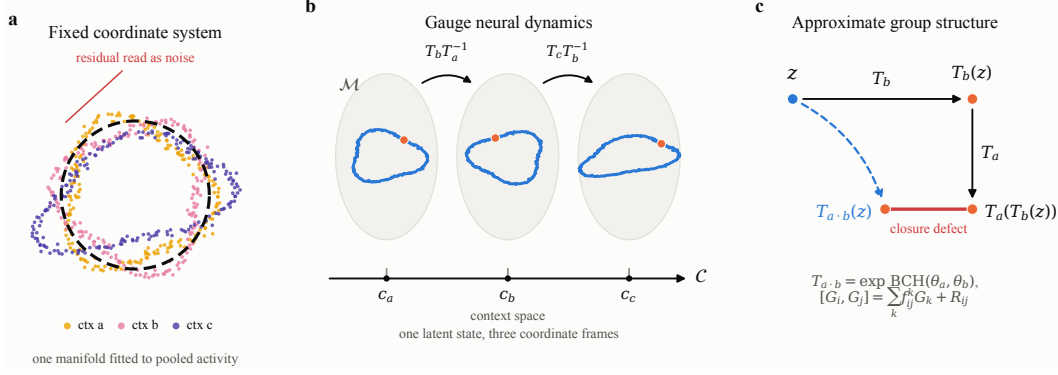


Figure 1: Gauge neural dynamics. (a) Fitting one manifold to pooled activity turns context structure into residual scatter. (b) Context space is the base of a bundle whose fibre is the latent manifold: one latent state, expressed in a context-dependent frame, with learned transitions between frames. (c) Composing the transformations of two contexts should land on the transformation of their composite, as predicted by BCH with the learned structure constants. The gap is the closure defect of Eq. (3).

3.4 Architecture and objective

The encoder E and decoder D are shared MLPs that never receive the context (Fig. 6, Appendix A): the encoder returns $w_c = E(x_c)$, the gauge pulls it back to $z = T_c^{-1}(w_c)$, and the decoder maps a frame-specific latent to activity. Giving the encoder the context would let it absorb the transformation and leave the gauge unidentifiable. The objective is

$$\begin{aligned} \mathcal{L} = & \underbrace{\|D(E(x_c)) - x_c\|^2}_{\text{reconstruction}} + \underbrace{\lambda_t \|D(T_b T_a^{-1} E(x_a)) - x_b\|^2}_{\text{transport}} + \underbrace{\lambda_i \mathbb{E}_s \text{tr Cov}_c[z_{s,c}]}_{\text{invariance}} \\ & + \underbrace{\lambda_g \frac{\|T_a T_b z - T_{a \cdot b} z\|^2}{\|T_a T_b z - z\|^2}}_{\text{group}} + \lambda_c \delta_{\text{closure}} + \lambda_h \mathcal{L}_{H_0}(x, z) + \text{reg}. \end{aligned} \quad (5)$$

The transport term does the work: predict the response in context b from one recorded in context a , using only the shared decoder and the learned transformation. The group term is normalised by how far the composite map moves points, which stops $T \equiv \text{id}$ from scoring perfectly, and $\mathcal{L}_{H_0}$ is the persistence-signature loss of Moor et al. (2020). The invariance term uses paired samples where correspondence is available; a correspondence-free variant replaces it with a multi-scale kernel MMD (Gretton et al., 2012) between canonical latent distributions. We report both.

4 Metrics

The metrics have to apply to methods that estimate no transformation at all, otherwise the comparison is not a comparison. For such a baseline we fit an affine map per context on the training split, take its real matrix logarithm, and extract a K -dimensional algebra by PCA; every metric then runs through the same code as for our model. This is generous, since it hands them the paired correspondence our correspondence-free variant does not use. All values are on held-out samples.

Gauge consistency (GCS) compares the realised composition against the BCH prediction, $\varepsilon(a, b) = \|T_a T_b z - T_{a \cdot b} z\| / \|T_a T_b z - z\|$, and reports $\max(0, 1 - \langle \varepsilon \rangle)$. The denominator is not decoration: without it a family collapsed onto the identity would score perfectly, and one of our baselines does exactly that. We therefore always report the transformation magnitude $\langle \|T_c(z) - z\| \rangle$ beside it.

Context invariance (CIS) is an intraclass correlation on the canonical latent, between/(between + within) with within $= \mathbb{E}_s \text{tr Cov}_c[z_{s,c}]$ and between $= \text{tr Cov}_s[\bar{z}_s]$. As the context component grows the raw ratio tends to $1/C$, not to zero, since averaging C contexts shrinks noise in the per-sample mean by that factor; we therefore chance-correct to $(\text{CIS} - 1/C)/(1 - 1/C)$, without which values would

not be comparable between five contexts and sixteen. A collapsed latent still scores zero, having no between-sample variance.

Geometric recovery (GRE) has to contend with the conjugation freedom. We fit one ridge-regularised affine chart ϕ from canonical latent to ground-truth coordinate on the reference context, freeze it, and measure the failure of $\phi \circ T_c = T_c^* \circ \phi$:

$$\text{GRE}_c = \|\phi(T_c(z)) - T_c^*(u)\| / \|T_c^*(u) - u\|. \quad (6)$$

Normalising by the size of the true transformation makes $\text{GRE} = 1$ the score of assuming no transformation, so only values below 1 mean anything. Where the ground-truth coordinate is not affine in the latent a fitted chart is unreliable; we say so rather than quoting the number, and fall back on a chart-free alternative, the leave-one-context-out R^2 of predicting $\text{vec log } T_c^*$ linearly from θ_c .

Manifold preservation (MPS) rests on manifolds related by a diffeomorphism having the same homology, so agreement of persistence diagrams across contexts is necessary for (1). We compute Vietoris–Rips diagrams of the canonical latent per context and report $1/(1 + \langle d_B \rangle)$, with d_B the bottleneck distance to the diagram of the true manifold. Bottleneck rather than Wasserstein, since the summed transport cost grows with the number of bars. Betti numbers come from the largest multiplicative gap in the barcode, reported with the achieved gap so a marginal call stays visible.

5 Experiments

Each simulation produces paired multi-context recordings: the same latent states seen through a shared bank of tuning curves in every context. Activity is z-scored with statistics pooled over contexts, since per-context scaling would remove part of what we are measuring. Models are fitted on 80% of samples and every number is on the remainder, as mean $\pm$ s.e.m. over five seeds. Baselines are PCA, UMAP (McInnes et al., 2018), an autoencoder, a VAE (Kingma and Welling, 2014), CCA, per-context PCA with orthogonal Procrustes, and semi-supervised manifold alignment (Appendix A).

5.1 Hippocampal remapping

A simulated rodent forages in a circular arena while 120 place cells tile it with Gaussian fields. Five contexts share one bank of tuning curves and differ only in the coordinate frame: the familiar environment, two rotations, an anisotropic stretch-and-shear, and a radially graded twist. The last of these is a diffeomorphism but not an affine map, and so lies outside the reach of the linear gauge by construction.

Fig. 2a shows what has to be explained: the field of one cell follows the cue. Fig. 2b shows the latent cloud with all contexts overlaid; before the gauge is applied the contexts fan apart, afterwards they collapse. Transport R^2 is 0.610 ± 0.000 for GND against 0.566 ± 0.000 for the best baseline (UMAP), with CIS 0.980 ± 0.000 and GRE 0.371 ± 0.000 .

Manifold alignment deserves a comment: it attains GCS 0.988 ± 0.000 , higher than our 0.891 ± 0.000 , while moving the latent by only 0.08 ± 0.00 against our 0.47 ± 0.00 . Its transformations compose because they are almost trivial, which is the failure mode the magnitude column exists to catch.

The morph context behaves as the geometry says it should (Table 6). It is a diffeomorphism but not an affine map, so the linear gauge cannot express it and scores 0.472 ± 0.000 there; the flow gauge, which extends the same construction to diffeomorphisms, reaches 0.408 ± 0.000 . Averaged over all contexts the two are closer, since four of the five are affine and the linear gauge handles those at least as well. The correspondence-free variant, which matches distributions instead of learning which samples correspond, reaches transport R^2 0.589 ± 0.000 and CIS 0.950 ± 0.000 ; that matters, since paired samples are rarely available in real recordings.

5.2 Grid-cell geometry

A single grid module’s activity depends on position only through a pair of phases, so its manifold is a torus (Gardner et al., 2022). Embedding the phase pair as $(\cos \theta_1, \sin \theta_1, \cos \theta_2, \sin \theta_2)/\sqrt{2}$ turns

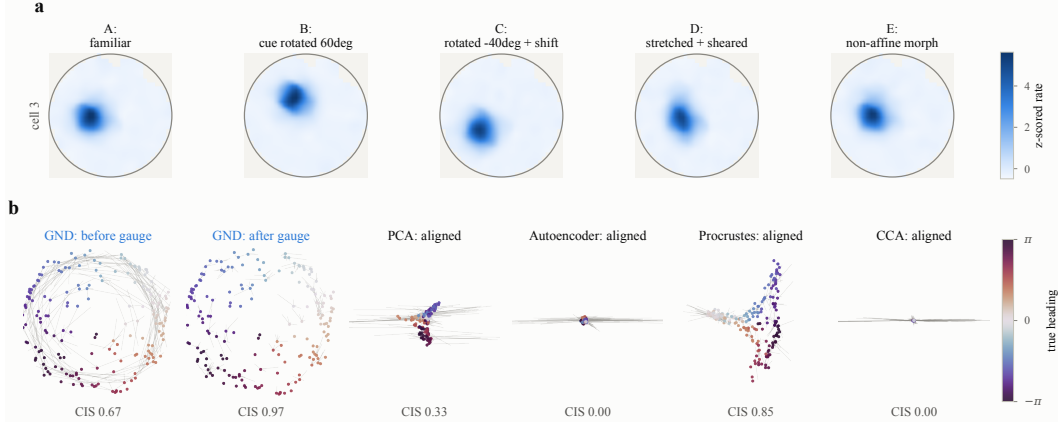


Figure 2: Hippocampal remapping. (a) Rate maps of one place cell across contexts; the field follows the cue coherently. (b) Latent cloud, all contexts overlaid, coloured by true heading. Aligned contexts give one smooth gradient; misaligned ones interleave. First panel: our observed-frame latent, before the gauge. Second: our canonical latent, after. Remainder: baselines after their own post-hoc alignment.

Table 1: Experiment 1: hippocampal place-cell remapping. Mean $\pm$ s.e.m. over 1 seeds, all on held-out samples. Transport R^2 is the accuracy of predicting the population response in another environment; GRE is normalised so that 1.0 is the score of assuming no transformation at all. The last column reports how far the recovered transformations actually move the latent, and must be read together with GCS: a family collapsed onto the identity composes perfectly while explaining nothing.

Method	Transport $R^2 \uparrow$	CIS $\uparrow$	GCS $\uparrow$	GRE $\downarrow$	MPS $\uparrow$	Alg. $R^2 \uparrow$	$\ T_c - \text{id}\ $
GND (ours)	0.610 ± 0.000	0.980 ± 0.000	0.891 ± 0.000	0.371 ± 0.000	0.890 ± 0.000	—	0.47 ± 0.00
GND-flow	0.680 ± 0.000	0.989 ± 0.000	0.869 ± 0.000	0.340 ± 0.000	0.916 ± 0.000	—	0.48 ± 0.00
GND-unpaired	0.589 ± 0.000	0.950 ± 0.000	0.885 ± 0.000	0.273 ± 0.000	0.857 ± 0.000	—	0.47 ± 0.00
PCA	0.497 ± 0.000	0.465 ± 0.000	0.873 ± 0.000	0.424 ± 0.000	0.867 ± 0.000	—	0.65 ± 0.00
UMAP	0.566 ± 0.000	0.946 ± 0.000	0.000 ± 0.000	0.243 ± 0.000	0.910 ± 0.000	—	0.50 ± 0.00
Autoencoder	0.347 ± 0.000	0.185 ± 0.000	0.881 ± 0.000	1.056 ± 0.000	0.829 ± 0.000	—	0.58 ± 0.00
VAE	0.307 ± 0.000	0.370 ± 0.000	0.889 ± 0.000	1.192 ± 0.000	0.816 ± 0.000	—	0.58 ± 0.00
CCA	-0.044 ± 0.000	0.172 ± 0.000	0.000 ± 0.000	2.570 ± 0.000	0.880 ± 0.000	—	1.23 ± 0.00
Procrustes	0.178 ± 0.000	0.877 ± 0.000	0.528 ± 0.000	3.014 ± 0.000	0.844 ± 0.000	—	1.13 ± 0.00
ManifoldAlign	0.211 ± 0.000	0.939 ± 0.000	0.988 ± 0.000	1.157 ± 0.000	0.852 ± 0.000	—	0.08 ± 0.00

a phase translation into a pair of independent plane rotations, an element of the maximal torus of $SO(4)$. Three predictions follow: the learned algebra should be abelian, composition should be exact already at BCH order one, and the canonical latent should have Betti numbers $(1, 2, 1)$.

Since a torus is not preserved by a general linear map, we restrict the gauge group to isometries; this is a geometric requirement rather than a convenience, and the unrestricted variant is worse (GRE 1.694 ± 0.000 against 0.447 ± 0.000 , Table 2, Appendix B). The predictions hold. The generators nearly commute, with mean commutator norm 0.451 ± 0.000 for unit-normalised generators, and GCS is 0.994 ± 0.000 . The canonical latent is a torus (Fig. 3c,d) that factorises into two circles carrying the two phases (Fig. 3e,f), and Betti numbers $(1, 2, 1)$ are recovered in 0% of runs.

Two further families probe the boundaries (Table 7). Adding the 60° lattice automorphism, an integer matrix and hence a genuine torus map, makes the group $T^2 \rtimes Z_6$ non-abelian, so a non-zero commutator is now the right answer and the norm rises to —; being non-linear on the $\mathbb{R}^4$ embedding, it defeats the linear gauge (GRE —) while the flow gauge does better (—). The third family is a control we expect to fail: a non-integer grid rescaling (Barry et al., 2007) is well defined on the universal cover but is not a map on the torus at all, so no latent gauge transformation can express it. GRE there is — and transport R^2 falls to —. The framework says in advance which phenomena lie outside it, and the measurements agree.

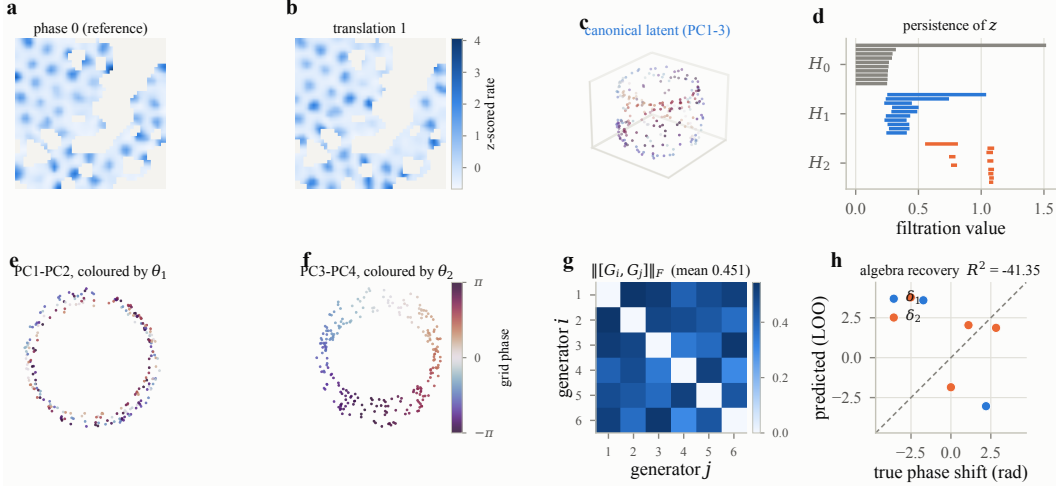


Figure 3: Grid-cell torus and translation group. (a,b) Spatial rate maps of one grid cell in the reference and a phase-shifted context. (c) Canonical latent in its leading three principal components. (d) Its persistence barcode. (e,f) The latent factorises into two circles carrying the two phases. (g) Pairwise commutator norms of the learned generators; for pure translations the correct algebra is abelian. (h) Leave-one-context-out prediction of the true phase shift from the learned algebra coordinates.

5.3 Motor cortex

A rate RNN is trained by BPTT to produce bell-shaped hand velocity toward each of eight targets at two reach extents. We impose no symmetry on the recurrent weights, so whatever equivariance the solution has is learned from the task. This lets us measure a ceiling instead of assuming an exact ground truth: after optimally aligning trial-averaged trajectories of equal-extent conditions, the circuit’s own equivariance residual is 0.206 ± 0.006 .

Contexts are the sixteen reach conditions. GND reaches transport R^2 0.876 ± 0.008 while every baseline is negative, the best being -0.103 ± 0.138 for Autoencoder. A negative value means the method predicts another condition’s population response worse than that condition’s own mean does. Trajectory alignment error is 0.053 ± 0.002 against 0.188 ± 0.009 for the best baseline (ManifoldAlign), and Fig. 4a,b shows the collapse directly. For the conjugacy prediction of (4), fitting linear dynamics separately per condition gives a frequency coefficient of variation of 0.332 ± 0.025 with mean rotation-plane angle 48.0 ± 4.1 degrees. The circuit’s own spread, measured from a PCA of the recorded activity, is 0.408 ± 0.037 , so the learned latent reproduces it to within 0.076.

Reading that comparison needs the same care as GCS. A small frequency CV is not in itself good: a latent that fails to separate conditions has no frequency spread at all. Manifold alignment reaches 0.066 ± 0.007 while its transport R^2 is -0.247 ± 0.115 . Matching the circuit’s value, not undercutting it, is what Eq. (4) predicts.

GRE is not a well-posed instrument here (2.457 ± 0.296). Commanded hand velocity is not an affine function of the learned latent in a context-independent way, so a chart fitted on the reference context extrapolates unreliably onto transported latents. That is a property of the measurement rather than of the model, and it is why we introduced the chart-free alternative and lead with transport and trajectory alignment for this experiment.

5.4 A field over context space

Discrete labels can only ask whether a transformation was fitted per recorded condition. The stronger question is whether the model learns a field over context space: a map $c \mapsto T_c$ correct at cue values never observed. We present thirteen cue orientations spanning a 120° arc, with the context variable being the cue direction $(\cos \alpha, \sin \alpha)$, and hold four out entirely.

The field interpolates (Fig. 5a,b; Table 4). At held-out cue angles transport R^2 is 0.688 ± 0.002 , essentially unchanged from 0.688 ± 0.002 at trained angles, against -0.024 ± 0.095 for leaving the

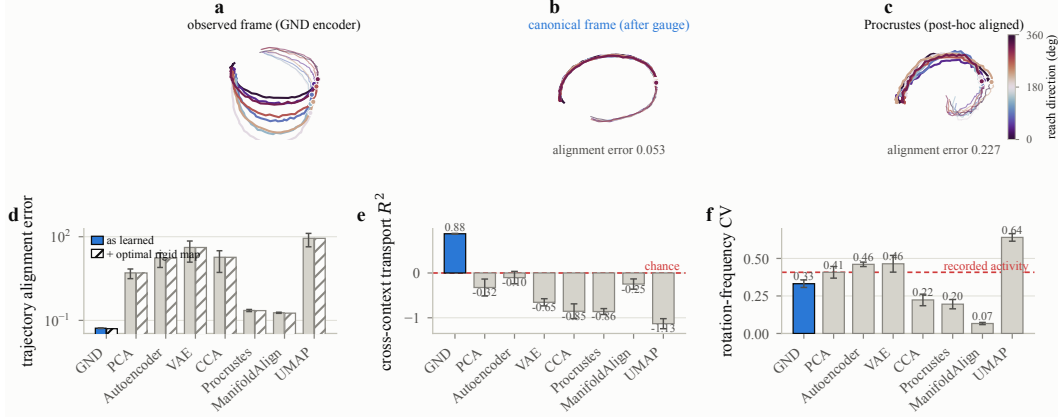


Figure 4: Motor-cortex reach conditions. (a) Trial-averaged trajectories in the observed frame, one colour per reach direction. (b) The same after inverting the learned gauge. (c) The strongest baseline after its own alignment. (d) Trajectory alignment error, with and without an additional optimal rigid map. (e) Cross-context transport. (f) Test of Eq. (4): spread of the top rotation frequency across conditions, with the value from recorded activity as reference.

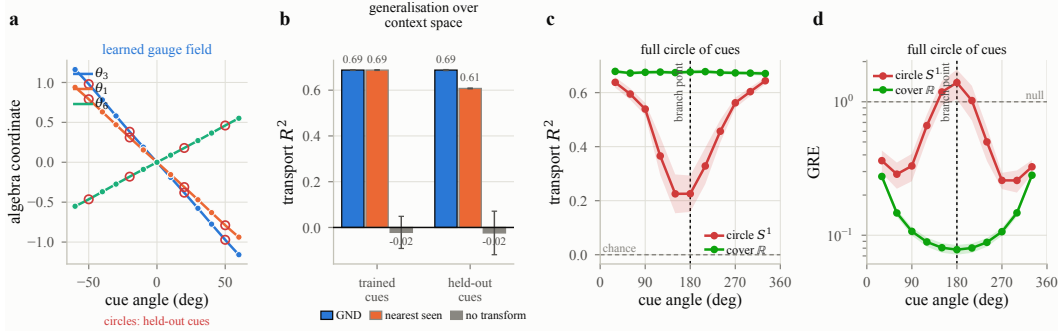


Figure 5: The gauge field over context space. (a) Learned algebra coordinates as a function of cue angle, queried on a dense grid; circles mark cues held out of training. (b) Transport at trained and held-out cue angles, against the only two baselines definable for an unobserved condition. (c,d) The winding obstruction. On a circular context space the failure is localised at the branch point antipodal to the reference cue; on the universal cover it disappears.

latent untransformed and 0.608 ± 0.002 for re-using the nearest observed angle’s transformation. Per-condition alignment methods cannot be evaluated here at all, having no transformation for a condition they never saw.

Closing the family into a full circle triggers Prop. 1. With context space S^1 , mean transport R^2 over the circle is 0.471 ± 0.024 , and the failure is localised: at the antipode of the reference cue it falls to 0.226 ± 0.067 with GRE 1.39 ± 0.35 , while near the reference it is unimpaired (Fig. 5c,d). Giving the identical architecture the universal cover, that is the unwrapped angle, removes the effect: 0.675 ± 0.002 on average and 0.676 ± 0.008 at 180° , with GRE 0.08 ± 0.01 . The two runs differ in nothing but the topology of the domain, which argues that this is the predicted obstruction rather than an optimisation artefact.

6 Ablations and robustness

Each ablation removes or replaces one ingredient and is refitted from scratch on the place-cell simulation (Fig. 7, Table 5, Appendix B). Removing the gauge, which is exactly the no-context latent model, costs -0.574 in transport R^2 : the largest single effect, and the clearest evidence that the gauge rather than the extra parameters is doing the work. Removing the group-consistency terms costs 0.004 in prediction while degrading GCS to 0.879 ± 0.000 , so a component can matter for geometry without mattering much for fit; removing the topology term costs 0.002 in transport but lowers MPS

from 0.887 ± 0.000 to 0.866 ± 0.000 . Truncating BCH at first order lowers GCS to $-$, as it must for a non-abelian family. The latent dimension sweep shows a useful asymmetry: under-parameterising is costly ($-$ at $d = 2$) while over-parameterising is nearly free ($-$ at $d = 20$), and since the true latent dimension of a recording is unknown, that is the asymmetry one wants.

For robustness (Fig. 8, Appendix B) we vary observation noise from 1 to 10 times the activity standard deviation, substitute Poisson spike counts, and vary population and sample size. Transport R^2 falls from 0.8 to 0.6 across the noise range while the best baseline reaches 0.7 ; under Poisson counts it is 0.7 . With recorded cells it is 0.7 , rising to 0.8 at 10 .

7 Discussion

Across three simulations drawn from different phenomena, treating context as a coordinate transformation of a shared latent rather than as extra latent variance recovers the generating transformation and predicts activity in contexts the model was never fitted on. The group structure is not decorative. The learned algebra is abelian exactly where the biology says it should be, for grid-cell phase translations, and non-abelian exactly where it should not be, once a lattice automorphism is added; and truncating the composition law degrades composition consistency as predicted.

Two predictions are concrete enough to take to data. The first is (4): if reach conditions are related by a latent gauge transformation, the eigenvalues of the condition-wise linearised dynamics, and in particular the rotation frequency, should be shared while the eigenplanes rotate. This is measurable in existing motor cortex datasets without fitting anything of ours. The second follows from Prop. 1: a smoothly varying cyclic context, such as a rotating cue, cannot be handled by any single continuous frame assignment, so remapping across a full 360° cue rotation should show a discontinuity or hysteresis somewhere on the circle, where a non-cyclic manipulation of comparable size should not.

These are simulations, built from published phenomenology and, for the motor case, from a trained rather than hand-designed network; nothing here is a claim about recorded data, and the next step is public grid-cell and motor cortex datasets. Several limitations are intrinsic. Rate remapping, in which firing rates change while fields stay put, is a change in f and not in T_c , and is outside (1) by construction, as is global remapping modelled as a permutation of cell identity, which is a gauge transformation in observation space. Grid rescaling is not a map on the population manifold at all. The chart dependence of GRE is a real weakness wherever the ground-truth coordinate is not affine in the latent. And the linear gauge assumes a global transformation; local deformations would need a position-dependent connection, of which the flow gauge is only a first step. Whether real neural systems maintain gauge structure of this kind is an empirical question, and we have tried to formulate it so that it can be answered rather than assumed.

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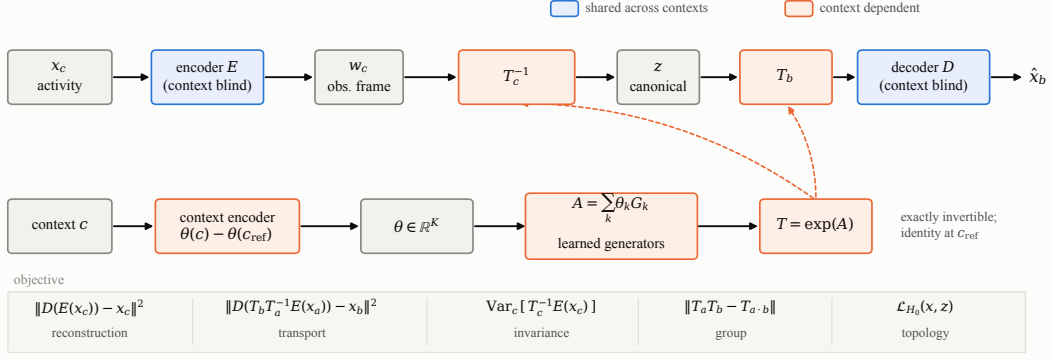


Figure 6: Architecture and objective. Only the gauge is context dependent. Encoder and decoder are shared and context blind, so a context difference has nowhere else to go.

A Implementation details

A.1 Compute and reproducibility

Everything runs on CPU. The models are small, under 10^6 parameters, and the gauge relies on `matrix_exp` and `linalg.lstsq`, neither of which is implemented for Apple’s MPS backend. Python, NumPy and PyTorch are seeded per run and deterministic kernels are requested. The five-seed suite takes a few hours on eight cores. A single entry point reproduces every number and figure from scratch; see the README.

A.2 Model and optimisation

Encoder and decoder are 3-layer MLPs, 192 hidden units, layer normalisation, GELU. The context encoder is a 2-layer tanh MLP with 64 hidden units and a small-scale output head, followed by anchoring at the reference context. The latent dimension is 6, except in the motor simulation where it is 8 to accommodate the condition-independent component, and $K = 6$ generators are used throughout. AdamW, learning rate 2×10^{-3} , weight decay 10^{-5} , batch size 256, gradient clipping at norm 1, cosine annealing over 200–250 epochs. The ablations use a shorter schedule of 120 epochs on 1500 samples, applied identically to every configuration including the unablated model, since what is being compared there is the effect of removing an ingredient rather than the absolute level. Loss weights are $\lambda_t = 1$, $\lambda_i = 1$, $\lambda_g = 0.2$, $\lambda_c = 0.05$, $\lambda_h = 0.5$. The gauge-dependent terms are ramped in linearly over the first 15% of training, so the encoder is not asked to satisfy a transformation before it has a latent to transform. Since θ depends only on the context, the matrix exponential is evaluated once per context per step rather than once per sample; transport and group terms are then batched over context pairs.

For the flow gauge, $V_k(z) = G_k z + v_k + \eta m_k(z)$ with $\eta = 0.5$ and m_k a 2-layer tanh MLP whose output layer is initialised to zero, so training starts exactly at the linear gauge. Flows use six RK4 steps. The vector-field bracket needs K^2 Jacobian-vector products, which dominates the cost, so structure constants are cached and refreshed every 25 steps and enter the objective as derived constants, in the spirit of a target network.

A.3 Simulations

Place cells. 120 cells with Gaussian fields on a sunflower tiling of a unit disc, width 0.22 with log-normal jitter, log-normal peak rates. Positions come from a momentum random walk with specular reflection at the wall, which gives the correlated, non-uniform occupancy of real foraging. Contexts act on position through I , $R(60^\circ)$, $R(-40^\circ)$ with a translation, an anisotropic stretch-and-shear $\text{diag}(1.25, 0.80)$ with shear 0.22, and a radially graded twist $u \mapsto R(0.9(\|u\|/R)^2)u$, which is a diffeomorphism of the disc but not affine.

Grid cells. 100 cells in one module, spacing 0.42, orientation 0.21 rad. Rates follow the standard three-cosine construction through an exponential non-linearity. Since $k_3 = k_2 - k_1$ for a hexagonal

lattice, the module’s activity depends on position only through the phase pair, which is what makes the manifold a torus; we report the homology of a single module, matching Gardner et al. (2022). Contexts act on the phase: translations by fixed, well-spread offsets; a 60° lattice rotation, whose matrix $KR_{60}K^{-1}$ is exactly integer and hence a genuine torus automorphism; and a non-integer rescaling by 1.37 as the control.

Motor cortex. A 128-unit continuous-time tanh rate RNN, $\tau\dot{x} = -x + W \tanh(x) + Bu(t) + b + \sqrt{2\tau\sigma}\xi(t)$, with $\tau = 50$ ms and $dt = 10$ ms, trained by BPTT for 1500 steps to emit hand velocity toward eight targets at two extents after a go cue, following a 300 ms preparatory period. Reaches are slightly curved, with lateral velocity equal to the derivative of the speed profile. The curvature is there for a reason. With perfectly straight reaches the canonical reach-plane coordinate is confined to a line, the second row of any readout from latent to reach plane is unconstrained by the reference context, and no method can be scored on recovering a rotation of that plane. Recorded activity is a random dense mixture of unit rates, as with an electrode array; twelve trials per condition are simulated with independent intrinsic noise. Splits are taken at the level of whole trials so no trajectory straddles the boundary.

A.4 Baselines

Each baseline supplies an observed-frame latent per sample per context and an estimated linear map from the canonical frame to each context’s frame; the canonical latent is $z_c = T_c^{-1}w_c$ and every metric applies unchanged. PCA, UMAP, the autoencoder and the VAE embed pooled activity and have their transformations fitted post hoc by least squares on the training split. Procrustes runs PCA per context and aligns to the reference by orthogonal Procrustes (Schönemann, 1966). CCA aligns each context to the reference using the paired correspondence explicitly. Manifold alignment builds a joint graph with within-context k -nearest-neighbour edges and across-context correspondence edges, embeds with the graph Laplacian (Ham et al., 2005), and extends out of sample by random-feature ridge regression.

For the headline transport score, every method including ours is read out through the same non-linear map: a random-Fourier-feature ridge regression (Rahimi and Recht, 2007) from observed-frame latent to activity, fitted on the training split. This separates the contribution of the latent and the transformation from decoder capacity. Where a method has a native decoder we report its own numbers too, and they are similar.

A.5 Metric details

Persistence diagrams come from `ripser` (Tralie et al., 2018) on subsamples of up to 500 points, with filtration values normalised by the mean pairwise distance so diagrams are comparable across embeddings of different scale. The essential class is given the diameter of the cloud as its death time; capping it at the largest finite death instead would make it indistinguishable from the longest noise bar and destroy the gap that Betti estimation relies on. Before comparing diagrams we drop bars shorter than 1% of the longest and keep at most the forty longest, which removes the near-diagonal noise band that would otherwise dominate an optimal matching. Betti numbers come from the largest multiplicative gap in the sorted lifetimes, required to exceed 1.5. That threshold is the one genuinely tuned constant in the analysis: at the sample sizes used here a disc produces spurious H_1 gaps of about 1.1 and a torus produces a true two-bar gap of about 1.9, so 1.5 separates them, but the separation for H_1 on a torus is the marginal case for any gap-based estimator and we report the achieved gap alongside the estimate.

The post-hoc algebra fitted to a baseline’s transformations uses the principal real matrix logarithm. We record the discarded imaginary part, which is large when a transformation lies outside the principal branch, for instance a rotation by more than π ; in that case the extracted algebra element is not unique. The GRE chart is ridge regularised with a relative penalty, which matters when the latent dimension exceeds that of the ground-truth coordinate. An unpenalised fit places weights of order $1/\sigma$ on low-variance latent directions, so the chart is accurate on the reference context but extrapolates wildly onto the transported latents it is then applied to, penalising a method for having a rich latent rather than for getting the transformation wrong.

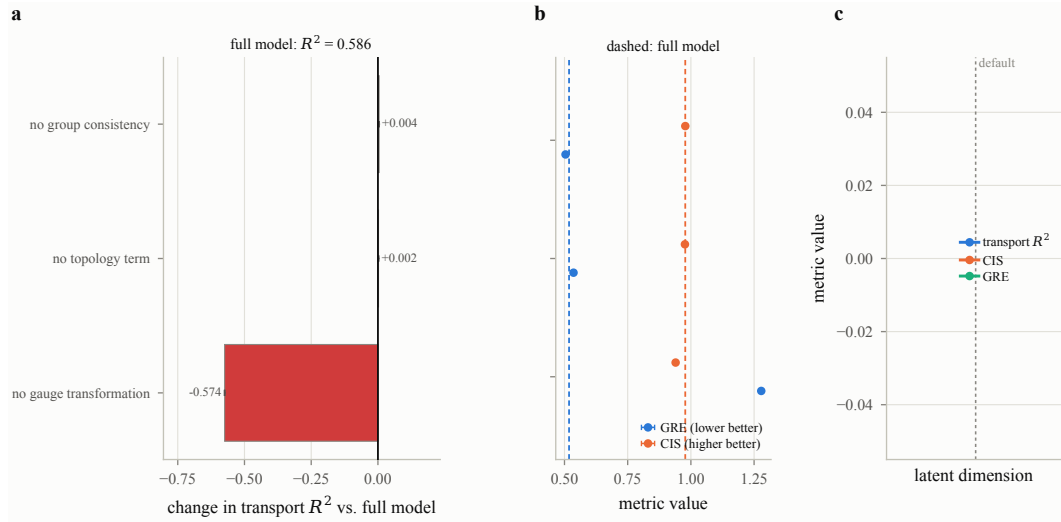


Figure 7: Ablations. (a) Change in transport R^2 when one ingredient is removed. (b) The same ablations on two geometric metrics, which respond differently. (c) Latent dimension sweep.

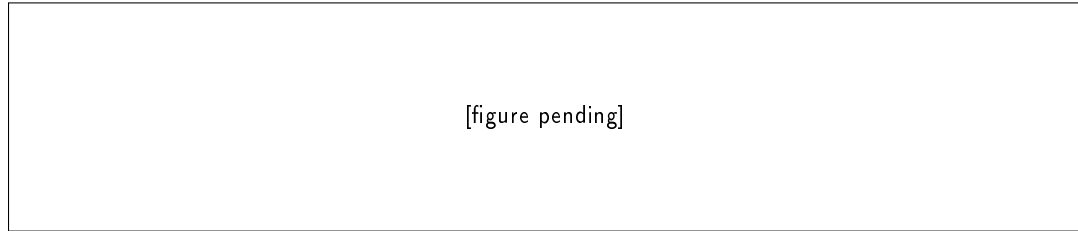


Figure 8: Robustness and scaling. (a,b) Observation noise. (c) Number of recorded cells. (d) Number of behavioural samples.

B Additional results

[table pending]

Table 2: Experiment 2: grid-cell phase translations. Mean $\pm$ s.e.m. over 1 seeds. The context family is the abelian translation group T^2 , which acts on the $\mathbb{R}^4$ torus embedding as a pair of independent plane rotations, so the correct algebra is commuting and the correct Betti numbers are (1, 2, 1).

Method	Transport $R^2 \uparrow$	CIS $\uparrow$	GCS $\uparrow$	GRE $\downarrow$	MPS $\uparrow$	Alg. $R^2 \uparrow$	$\ T_c - \text{id}\ $
GND (ours)	0.941 $\pm$ 0.000	0.999 $\pm$ 0.000	0.994 $\pm$ 0.000	0.447 $\pm$ 0.000	0.901 $\pm$ 0.000	—	1.18 $\pm$ 0.00
GND-flow	-1.178 $\pm$ 0.000	0.981 $\pm$ 0.000	0.941 $\pm$ 0.000	10.276 $\pm$ 0.000	0.820 $\pm$ 0.000	—	0.96 $\pm$ 0.00
GND-gl	-0.299 $\pm$ 0.000	0.982 $\pm$ 0.000	0.934 $\pm$ 0.000	1.694 $\pm$ 0.000	0.814 $\pm$ 0.000	—	1.16 $\pm$ 0.00
PCA	0.847 $\pm$ 0.000	0.936 $\pm$ 0.000	0.905 $\pm$ 0.000	0.158 $\pm$ 0.000	0.880 $\pm$ 0.000	—	1.21 $\pm$ 0.00
UMAP	0.226 $\pm$ 0.000	0.246 $\pm$ 0.000	0.580 $\pm$ 0.000	0.553 $\pm$ 0.000	0.852 $\pm$ 0.000	—	1.05 $\pm$ 0.00
Autoencoder	0.801 $\pm$ 0.000	0.903 $\pm$ 0.000	0.480 $\pm$ 0.000	0.217 $\pm$ 0.000	0.910 $\pm$ 0.000	—	1.20 $\pm$ 0.00
VAE	0.117 $\pm$ 0.000	0.722 $\pm$ 0.000	0.000 $\pm$ 0.000	0.441 $\pm$ 0.000	0.854 $\pm$ 0.000	—	1.00 $\pm$ 0.00
CCA	-0.184 $\pm$ 0.000	0.717 $\pm$ 0.000	0.488 $\pm$ 0.000	1.127 $\pm$ 0.000	0.827 $\pm$ 0.000	—	1.37 $\pm$ 0.00
Procrustes	0.152 $\pm$ 0.000	0.931 $\pm$ 0.000	0.601 $\pm$ 0.000	0.725 $\pm$ 0.000	0.868 $\pm$ 0.000	—	1.05 $\pm$ 0.00
ManifoldAlign	0.112 $\pm$ 0.000	0.976 $\pm$ 0.000	0.992 $\pm$ 0.000	0.974 $\pm$ 0.000	0.912 $\pm$ 0.000	—	0.04 $\pm$ 0.00

Table 3: Experiment 3: motor-cortex reach conditions. Mean $\pm$ s.e.m. over 5 seeds. Trajectory error is the residual between trial-averaged canonical trajectories of different reach conditions. Frequency CV tests Eq. (4): conjugation preserves eigenvalues, so rotation frequency should be shared across conditions while the rotation plane turns.

Method	Transport $R^2 \uparrow$	Traj. err. $\downarrow$	CIS $\uparrow$	GCS $\uparrow$	Freq. CV $\downarrow$	Plane angle (deg)
GND (ours)	0.876 $\pm$ 0.008	0.053 $\pm$ 0.002	0.998 $\pm$ 0.000	0.971 $\pm$ 0.002	0.332 $\pm$ 0.025	48.0 $\pm$ 4.1
PCA	-0.324 $\pm$ 0.189	5.053 $\pm$ 1.916	0.033 $\pm$ 0.020	0.072 $\pm$ 0.072	0.408 $\pm$ 0.039	61.3 $\pm$ 2.1
UMAP	-1.132 $\pm$ 0.112	87.521 $\pm$ 44.031	0.003 $\pm$ 0.003	0.000 $\pm$ 0.000	0.640 $\pm$ 0.025	62.1 $\pm$ 1.4
Autoencoder	-0.103 $\pm$ 0.138	17.175 $\pm$ 9.277	0.005 $\pm$ 0.003	0.000 $\pm$ 0.000	0.461 $\pm$ 0.015	60.8 $\pm$ 2.1
VAE	-0.654 $\pm$ 0.080	41.181 $\pm$ 28.948	0.010 $\pm$ 0.005	0.000 $\pm$ 0.000	0.464 $\pm$ 0.056	55.1 $\pm$ 2.4
CCA	-0.855 $\pm$ 0.166	18.344 $\pm$ 13.102	0.312 $\pm$ 0.119	0.440 $\pm$ 0.137	0.223 $\pm$ 0.038	63.1 $\pm$ 2.0
Procrustes	-0.860 $\pm$ 0.066	0.227 $\pm$ 0.020	0.967 $\pm$ 0.003	0.478 $\pm$ 0.080	0.195 $\pm$ 0.031	61.7 $\pm$ 1.4
ManifoldAlign	-0.247 $\pm$ 0.115	0.188 $\pm$ 0.009	0.968 $\pm$ 0.003	0.974 $\pm$ 0.002	0.066 $\pm$ 0.007	15.2 $\pm$ 1.9

Table 4: Generalisation over context space, and the winding obstruction. Top: cue angles held out entirely from training are predicted from the cue value alone. Bottom: when the cue family closes into a circle, a single-chart field cannot represent the required winding; giving the same architecture the universal cover removes the problem. Mean $\pm$ s.e.m. over 5 seeds.

Setting	Transport $R^2 \uparrow$	GRE $\downarrow$	Transport R^2 , no transform
trained cue angles	0.688 $\pm$ 0.002	0.308 $\pm$ 0.045	-0.022 $\pm$ 0.070
held-out cue angles	0.688 $\pm$ 0.002	0.247 $\pm$ 0.035	-0.024 $\pm$ 0.095
full circle, context space = S^1	0.471 $\pm$ 0.024	0.60 $\pm$ 0.07	—
full circle, universal cover $\mathbb{R}$	0.675 $\pm$ 0.002	0.13 $\pm$ 0.01	—

Table 5: Ablations. Each row removes or replaces one ingredient and is refitted from scratch on the place-cell simulation; Δ is the change in transport R^2 relative to the full model. Mean $\pm$ s.e.m. over 1 seeds. The latent-dimension sweep is shown in Fig. 7c.

Configuration	Transport R^2	Δ	CIS	GCS	GRE
full model	0.586 $\pm$ 0.000	—	0.978 $\pm$ 0.000	0.890 $\pm$ 0.000	0.518 $\pm$ 0.000
A1: no gauge	0.012 $\pm$ 0.000	-0.574	0.940 $\pm$ 0.000	1.000 $\pm$ 0.000	1.278 $\pm$ 0.000
A2: no group loss	0.591 $\pm$ 0.000	+0.004	0.978 $\pm$ 0.000	0.879 $\pm$ 0.000	0.504 $\pm$ 0.000
A2a: no closure only	0.587 $\pm$ 0.000	+0.001	0.978 $\pm$ 0.000	0.890 $\pm$ 0.000	0.518 $\pm$ 0.000
A3: no topology	0.588 $\pm$ 0.000	+0.002	0.976 $\pm$ 0.000	0.901 $\pm$ 0.000	0.536 $\pm$ 0.000

Table 6: Per-context recovery error in Experiment 1. The final context is a radially graded twist, which is a diffeomorphism but not an affine map, and so lies outside the reach of the linear gauge by construction; the flow gauge recovers it.

Model	B: cue rotated 60deg	C: rotated -40deg + shift	D: stretched + sheared	E: non-affine morph
GND	0.17	0.21	0.63	0.47
GND-flow	0.12	0.16	0.67	0.41
PCA	0.22	0.31	0.69	0.47

Table 7: Group structure across grid-cell context families. Adding the 60° lattice automorphism makes the group non-abelian, so a non-zero commutator is then the correct answer. The rescaling family is a negative control: a non-integer gain change is well defined on the universal cover but not on the torus, so no latent gauge transformation of the population manifold can express it.

Context family	Model	GCS $\uparrow$	$\ [\overline{G_i}, \overline{G_j}]\ $	GRE $\downarrow$	Transport $R^2 \uparrow$
T^2 (abelian)	GND	0.994 ± 0.000	0.451 ± 0.000	0.45 ± 0.00	0.941 ± 0.000
	GND-gl	0.934 ± 0.000	0.556 ± 0.000	1.69 ± 0.00	-0.299 ± 0.000
	GND-flow	0.941 ± 0.000	0.808 ± 0.000	10.28 ± 0.00	-1.178 ± 0.000